

CT-Question

1. True/ False

1*6

- a. DNS always uses UDP for all types of queries.
 - b. In non-persistent HTTP, multiple objects **cannot** be downloaded in parallel even using multiple TCP connections.
 - c. HTTP/1.1 persistent connections reduce both RTT overhead *and* server load.
 - d. A CDN (Content Delivery Network) eliminates the need for the origin server entirely.
 - e. DNS caching can introduce consistency problems if records change before TTL expiration.
 - f. TCP's congestion control can reduce video streaming quality in DASH-style adaptive systems.
2. Suppose that your department has a local DNS server for all computers in the department. You are an ordinary user (i.e., not a network/system administrator). Can you determine if an external Web site was likely accessed from a computer in your department a couple of seconds ago? Explain.
3. Why does persistent HTTP reduce delay compared to non-persistent HTTP?
4. What problem do cookies solve in HTTP?

1*3

1*3

1*3

CT-2

1. True/ False

- a. TCP uses a 4-tuple (source IP, destination IP, source port, destination port) for demultiplexing. 1*7
- b. In rdt2.0, the protocol correctly handles corrupted ACK/NAK messages.
- c. rdt3.0 introduces a timer to deal with lost packets.
- d. Stop-and-wait protocols achieve high link utilization on long-delay high-bandwidth networks.
- e. In Go-Back-N, the receiver must buffer out-of-order packets.
- f. In ad-hoc wireless networks, nodes communicate only through a base station.
- g. CDMA allows multiple users to transmit simultaneously using orthogonal codes.

2. Describe the main problem with rdt2.0 that leads to the design of rdt2.1 and rdt2.2. Why are sequence numbers essential? 1*5

3. Explain why stop-and-wait (rdt3.0) suffers from poor utilization in high-bandwidth, long-delay networks. Provide an example scenario. 1*4

4. Explain the hidden terminal problem in wireless networks and describe why it does not occur in wired Ethernet. 1*4

1. True/False

- a. In infrastructure mode, handoff occurs when a mobile node changes its IP address.
- b. In CDMA, perfect orthogonality of codes guarantees zero interference regardless of synchronization.
- c. 802.11 does not support collision detection because of the large difference between transmit and receive signal power.
- d. A known-plaintext attack is generally more powerful than a ciphertext-only attack.
- e. Digital signatures typically encrypt the entire message using the sender's private key.
- f. TLS uses public-key cryptography only during the handshake phase.

2. What is meant by "hand over" of a network device? Analyze the handover mechanism in a 4G/5G network by identifying the network elements involved in handover initiation and target base-station selection and explain how the process is performed?

4

3. Consider RSA with $p = 7$ and $q = 13$.

4

- e) What are n and z ?
- f) Let e be 17. Why is this an acceptable choice for e ?
- g) Find d such that $de = 1 \pmod{z}$.
- h) Encrypt the message $m = 9$ using the key (n, e) . Let c denote the corresponding ciphertext. Show all work.

4. Does the LTE wireless access network use FDMA, TDMA, or both? Justify your answer. Suppose Bob initiates a TCP connection to Trudy who is pretending to be Alice. During the handshake, Trudy sends Bob Alice's certificate. In what step of the TLS handshake algorithm will Bob discover that he is not communicating with Alice?